

DATE: .../.../20...

M T W T F S S

Part-paper-2017 Short

Q. (i) Difference between data mining and data ware housing?
Solved

(ii) What is noise?

(iii) What is classification?

(iv) What is regression?

(v) What is the significance of apriori property?

(vi) What is market basket analysis?

(vii) What is frequent pattern in association mining?

(ix) Why do we preprocess data?

Q(x) Discuss application area of data mining?

- o Research
- o Education sector
- o Transportation
- o market basket analysis
- o Intrusion detection
- o Scientific analysis
- o Health care

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Q. What is predictive data mining?
Type of data mining that focuses on historical data to make predictions about future events or outcomes. It involves identifying patterns and trends in existing data and then creating models that can forecast future behavior or results.

e.g. for example, a store can use past purchase data to predict what products a customer might buy in the future.

Q. What do you mean by selection in data mining?

o Refers to the process of choosing the relevant subset of data from a large dataset that will be used for analysis.

o Involves identifying and extracting data that will solve

problem. Improve efficiency & accuracy of data mining results.

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MINI PROJECT

Why we focus on dimensionality reduction in data mining?

⇒ Dimensionality reduction makes data mining more efficient & effective by:

- Simplifying data
- Improving algorithm performance
- Enhancing visualization
- Reducing noise
- Preventing overfitting
- Reducing storage

What is closed pattern?

⇒ refers to a frequent itemset that has no superset (a large group of items that includes the original group) with the same frequency.

⇒ offers a more concise way to represent and analyze frequent item combinations.

⇒ Provide efficient and compact representation of data.

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MINI PROJECT

What is maximal pattern?

set of items that appear together frequently, and it cannot be extended by adding more items while maintaining the same frequency.

TID	Items	
T ₁	bread, milk	→ maximal Pattern
T ₂	bread, milk, butter	
T ₃	milk, butter	← frequent items
T ₄	bread, milk, butter	

closed Pattern
b.c. no large item appears with same frequency

Q. Give an example of discrete data in data mining?

- Binary data
- categorical
- Count data
- ordinal
- Nominal

What is MODEL in data mining?

⇒ A model is a representation of patterns, relationships or structures discovered from the data.

⇒ a mathematical model or computational framework that captures the underlying characteristics of the data, allowing us to make predictions, classification or other types of analyses.

⇒ Types of model

- Regression
- Classification
- Clustering
- Association
- Neural network
- Decision Tree

List types of data mining?